

Rinku Diwakar

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TECHNICAL SKILLS

Machine Learning & MLOps: Scikit-learn, XGBoost, LightGBM, CatBoost, Feature Engineering, Feature Selection, Model Evaluation, Hyperparameter Tuning, GridSearchCV, RandomSearchCV, Regression, Classification, Clustering, Dimensionality Reduction (PCA, t-SNE), Anomaly Detection, Time Series Forecasting (ARIMA, SARIMA, Prophet), Ensemble Methods (Bagging, Boosting, Stacking), DVC, MLflow, CI/CD, Model Deployment, Model Monitoring

Deep Learning: CNN, LSTM, RNN, GRU, Transformer Architecture, Attention Mechanisms, Transfer Learning, Fine-Tuning, Object Detection (YOLO), Image Segmentation, NLP (Tokenization, Embeddings, Sentiment Analysis), OpenCV, Neural Networks, Backpropagation, Dropout, Batch Normalization

Generative AI & LLMs: Prompt Engineering, RAG (Retrieval-Augmented Generation), LangChain, Hugging Face Transformers, OpenAI API, Vector Databases, Embeddings, Fine-Tuning LLMs

Containerization & Cloud: Docker, Kubernetes (EKS), AWS (EC2, S3, ECR, IAM, CloudFormation, Auto Scaling), CI/CD Pipelines (GitHub Actions), Dagshub, MongoDB, Firebase, Supabase, Vercel

Programming: Python, C/C++, JavaScript, OOP, Data Structures & Algorithms, Git, REST APIs

Data Analytics & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, MySQL, Power BI (DAX), Advanced Excel, EDA, Statistical Analysis

Web Development: React.js, Flask API, Gunicorn, HTML5, CSS3, REST APIs, Responsive Design

Model Serving & Observability: MLflow Model Registry, Prometheus, Grafana, Health/Readiness Endpoints, Metrics Scraping, Artifact Management

DevOps & Infrastructure: eksctl, kubectl, AWS EKS, EC2 (Ubuntu), Auto Scaling Groups, Load Balancer, Security Groups, CloudFormation Stacks, PersistentVolumeClaims (PVC), DVC (Data Version Control), pipreqs, conda

Core Skills: Technical Leadership, Problem Solving, Business Strategy

LeetCode: Active problem solver focused on Data Structures & Algorithms



EDUCATION

- **Dr. B R Ambedkar National Institute of Technology (NIT) Jalandhar**
Bachelor of Technology – Electrical Engineering (CGPA: 7.54) 2023 – 2027
- **MGM KPS, Bheeti Kaushambi**
Class XII (CBSE) – 90% 2022 – 2023
- **MGM KPS, Bheeti Kaushambi**
Class X (CBSE) – 98.4% 2020 – 2021

PROFESSIONAL EXPERIENCE

Data Analytics & ML Intern | TS Bridge | Remote June 2025 – July 2025

- Developed end-to-end ML applications including bike-demand prediction using XGBoost and Scikit-learn; deployed via Flask API with JavaScript dashboard
- Executed data preprocessing, EDA, and model evaluation using Pandas, NumPy, SQL, Matplotlib, and Seaborn
- Implemented MLOps workflows with DVC and MLflow; built CI/CD pipelines and containerized models using Docker
- Assisted in cloud deployment and backend integration using AWS (EC2, S3) and MongoDB
- Built Power BI dashboards with DAX for KPI tracking and business insights

Data Science Intern | CourseVita | Remote June 2024 – July 2024

- Led data preprocessing project for machine learning dataset preparation and cleaning
- Performed data wrangling including missing values handling and duplicate removal using Python (Pandas, NumPy)
- Conducted exploratory data analysis to ensure data integrity and delivered clean dataset for predictive modeling

TECHNICAL PROJECTS

- **MovieSentiment – Full-Stack ML Movie Review Platform (Capstone)** 
Tech: Python, React.js, Flask, Gunicorn, Scikit-learn, MLflow, DVC, Dagshub, Docker, Kubernetes (EKS), Supabase, AWS (EC2, S3, ECR, IAM, CloudFormation), GitHub Actions, Prometheus, Grafana, eksctl, kubectl, Cookiecutter Data Science

Built a production-ready full-stack ML platform with sentiment analysis (positive/negative classification on IMDB data). Designed complete MLOps pipeline: experiment tracking on Dagshub/MLflow, data versioning with DVC (S3 remote), multi-stage Docker build serving React + Flask/Gunicorn from a single container. Deployed to Amazon EKS via GitHub

Actions CI/CD (ECR image push → kubectl apply) with HPA, health/readiness probes, and Load Balancer. Set up Prometheus (EC2) scraping app metrics and Grafana dashboards for real-time observability. Integrated WatchMode API for movie metadata and Supabase for reviews, ratings, and aggregate statistics.

- **Vehicle Insurance Domain Project (End-to-End ML + MLOps)**



Tech: Python, Scikit-Learn, DVC, MLflow, Docker, MongoDB, CI/CD, AWS

Built production-ready ML pipeline with data versioning, experiment tracking, containerization, and cloud deployment.

- **Kavach – Voice-Based Smart Security System**



Tech: Python, Vosk, Resemblyzer, Raspberry Pi, Arduino, REST APIs

Developed voice authentication system using Vosk (speech recognition) and Resemblyzer (speaker verification); deployed backend server on Raspberry Pi and integrated with Arduino-based hardware prototype for real-time door access control.

- **Seoul Bike Sharing Demand Prediction (ML)**



Tech: Python, Pandas, Scikit-Learn, XGBoost, Flask, JavaScript

Engineered regression models with 20+ features; achieved 15% RMSE reduction via GridSearchCV and deployed using Flask API.

- **MessOS – Smart Mess Management System**



Tech: React.js, Supabase, JavaScript, Database Management

Built full-stack hostel mess platform with student dashboard, meal tracking, and backend integration.

LEADERSHIP & CAMPUS INVOLVEMENT (NIT JALANDHAR)

- **Vice President, SEED Society** – Leading 30+ members; overseeing execution of technical initiatives and team operations
- **Internship Representative, Department of Electrical Engineering** – Coordinating between students and administration for internship opportunities and communication
- **Event Coordinator, Forge 1.0** – Directed end-to-end execution of innovation event with 200+ participants; managed sponsorship, PR, and logistics
- Coordinated cross-functional efforts across 4+ organizations including IIC, Robotics Club, E-Cell, and SEED Society

ACHIEVEMENTS

- Winner, ImaGenAI Innovation Challenge at **IIT Delhi Techfest 2024**
- **3rd Place**, What If Ideation Competition at **IIT Ropar Techfest 2024**
- Participant, Hover Hurdle Robotics Competition at **IIT Ropar & IIT Delhi Techfest**
- **Recognized for operational excellence** in customer support with 95% satisfaction rating
- **Successfully organized 10+ events** with 200+ participants, managing logistics, budgets, and vendor coordination